Auditing Context and Evaluation in a Powered Cube Draft Recommender

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Abstract

Powered Cube drafting requires selecting cards whose joint contribution can exceed their standalone value. We audit a Magic: The Gathering Arena recommender that combines contextual pick imitation with outcome-derived preference learning. Historical experiments improve agreement with generated preferences, but do not establish better drafting or gameplay. We identify a context-independent supervision path, a mismatch between partial draft pools and completed-deck value estimation, and defects in live state and gameplay evidence processing. After repairing the correctness paths, we construct a versioned queue of 150 cases: 24 authored contextual probes and 126 logged decisions. We replay only its 100-case development split. On 82 logged decisions, the raw policy matches 29 human choices and an aggressive preference-trained variant matches 21. On nine paired probes, their score changes follow the proposed direction in five and four cases, respectively. These are unreviewed development diagnostics, not expert accuracy or win-rate estimates. We provide reproducible evaluation artifacts and a package-aware redesign specification. Package-aware learning and prospective gameplay validation remain future work.

1. Introduction

A draft recommender must help assemble a functional deck over a sequence of constrained choices. A card that is weak without support may become a key pick once its partners are present. Conversely, a powerful payoff can be redundant when the pool lacks the means to use it. In Powered Cube, fast mana, tutors, graveyard strategies, and other interactions make this distinction consequential. The official Arena guide describes both standalone power and combinations, and supports hybrid strategies rather than requiring a fixed archetype commitment (Barash, 2026).

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The system studied here already conditions a neural policy on previously selected and seen cards. Nevertheless, its later training and inference layers can favor standalone card effects. This distinction matters: adding contextual inputs does not ensure that the supervision rewards contextual decisions. Nor does predicting the outcome of a completed deck automatically provide the marginal value of a card in an incomplete draft.

This paper documents a system audit and an evaluation revision, rather than presenting a newly validated combo-aware policy. Our contributions are: (1) a code-grounded account of the policy, value, and preference pipeline; (2) identification and repair of defects affecting draft state, card coverage, and outcome evidence; and (3) a versioned development evaluation that distinguishes imitation, teacher agreement, contextual sensitivity, and gameplay performance. The resulting design specification focuses on attainable decks and compatible card packages, but its proposed learning and deckbuilding components have not yet been implemented or evaluated.

2. Task and Related Work

We model an eight-seat draft with three packs of fifteen cards per seat. At pick t, the recommender receives

$$s_t = (x_t, y_t, z_t, m, p_t, k_t), \tag{1}$$

where x_t identifies cards in the current pack, y_t is the drafted pool, z_t counts previously seen but unpicked cards, m identifies the cube list, and (p_t, k_t) gives draft position. The legal actions are the cards in x_t. Seen counts may include repeated sightings; they are not counts of distinct remaining cards.

The desired quantity is an action's expected contribution to the eventual playable deck:

$$Q^*(s_t, a) = \mathbb{E}[U(B^*(Y_T)) \mid s_t, a], \tag{2}$$

where Y_T is the final pool, B^* denotes an appropriate deckbuilding procedure, and U is performance under a specified gameplay environment. Neither Q^* nor the counterfactual outcome of an unpicked card is directly observed in a draft log. Player skill, later picks, deck construction, and opponents all affect the realized result.

Contextual preference ranking has previously been applied to Magic drafting (Bertram et al., 2021). Our policy uses permutation-invariant set encoders related to Deep Sets (Zaheer et al., 2017), followed by candidate scoring. We use direct preference optimization (DPO) as a preference-fitting method (Rafailov et al., 2023); its objective does not validate the preferences supplied to it. Our contribution is an audit of this particular composition and its evidence, not a claim to introduce contextual drafting or set encoding.

3. Implemented System

3.1. Data and card representations

The historical datasets are local releases named `17lands_cube_powered_2025` and corresponding weighted and outcome datasets. Their manifests identify public 17Lands Powered Cube draft and game data and include Premier and Traditional Draft (17Lands, 2026). Pick and outcome ETL split by draft identifier. This prevents picks from one draft crossing a given split, but does not by itself address repeated model selection on that evaluation set, player overlap, or temporal shift. Replay data and turn-level gameplay skill modeling are not part of the evaluated training pipeline.

Each card has representation

$$e_i = E_i + \alpha g(f_i). \tag{3}$$

Here E_i is a learned embedding, and f_i contains structured card attributes: mana value, colors and color identity, types, power/toughness, rarity, and selected keywords. The local card cache is derived from Scryfall metadata (Scryfall, 2026). The main policy's feature vector does *not* include a general oracle-text embedding.

Rules text is used separately in the nearest-neighbor fallback. For a card outside the trained embedding table, the policy can replace its learned component with

$$\widehat{E}_i = \sum_{j \in \mathcal{N}_k(i)} w_{ij} E_j. \tag{4}$$

The neighbor space combines structured attributes, hashed rules text, and hand-specified functional tags. The current assistant uses up to eight neighbors, temperature 0.15, and an available cache; uncached targets use feature-based neighbor search. This is an inference fallback, not evidence that the model has learned a new card's combinations.

3.2. Contextual pick policy

Separate weighted-mean set encoders summarize the pool, seen cards, and cube. For example,

$$h_y = \rho_y \left(\frac{\sum_i y_i \phi_y(e_i)}{\max(1, \sum_i y_i)} \right). \tag{5}$$

A context MLP combines these summaries with color commitment and draft-position embeddings. Candidate logits are

$$\ell_i(s) = h(s)^\top e_i + b(e_i), \tag{6}$$

with a masked softmax over the pack. Training supports human-pick cross-entropy and a margin-ranking term against unpicked candidates. Outcome-weighted continuation is also implemented. Softmax scores are not established to be calibrated probabilities of pick quality.

These representations permit contextual dependence, but compress the pool before scoring. They do not explicitly track missing package roles, target restrictions, or multi-card execution requirements. Our experiments do not isolate this architectural choice from the effects of the training labels.

3.3. Value model and actual inference score

A separate model encodes a main deck, sideboard, and covariates to predict an outcome probability V_ψ. Its aggregate training uses game-count-weighted binary cross-entropy against each build's empirical success rate. A heuristic builder B maps a partial pool to the model's input by selecting spells, including nonbasic lands, and adding basics.

The implemented combined score uses policy logits and a *probability*-valued term:

$$\widetilde{V}_T(y) = \sigma(\mathrm{logit}(V_\psi(B(y)))/T), \tag{7}$$

$$S(a \mid s) = \ell_a(s) + \lambda \widetilde{V}_T(y + \{a\}) + \eta_t H(y, a) + P(a) + C(y, a). \tag{8}$$

H is the optional heuristic synergy score, P is a configured card prior, and C is a companion-constraint adjustment. The value term is not a value logit added to the policy. The current default sets $\lambda = \eta_t = 0$, while retaining the configured prior and companion rules. Experimental presets can enable the older value path.

The builder is a lightweight heuristic, not a validated constrained optimizer. It selects up to 23 nonland spells using coarse color and spell priorities, includes nonbasics, ignores cards absent from the value vocabulary, and fills toward forty cards with basics. Thus an early hypothetical pool can become one spell and 39 basics. A value model trained on completed decks is then evaluated far from its training distribution. There is no demonstrated rank marginalization, curve feasibility guarantee, or calibrated counterfactual interpretation for this score.

3.4. Preference learning

Historical preference datasets come from two different teachers: completed-deck value estimates produced through the builder, and a regularized card-main-effect outcome

model. The latter takes the form

$$\operatorname{logit} \Pr(w = 1) = \beta_0 + \sum_i d_i \beta_i + \gamma^\top r. \tag{9}$$

With pair effects disabled, the generated preference between two pack cards depends on their learned card coefficients, not their interaction with the current pool. DPO fits these preferences relative to a reference policy, with an optional explicit KL penalty. A contextual student can therefore be trained toward a context-independent ordering. This is a limitation of the teacher signal; it does not establish that every student decision is context-free.

4. Correctness and Evidence Audit

Live state and scoring parity. The audit found that changing draft identity did not clear pool and seen state, and that kNN fallback was triggered by enlarging the configured cube rather than by embedding-vocabulary coverage. Repairs reset context and pack deduplication on a new draft, initialize fallback representations for new cards already in the configured list, and share one loader/scorer between live and one-shot recommendation. A related value-model embedding-range crash now falls back to its existing unknown row. The live default again selects the outcome-weighted policy; an explicit aggressive preset remains available. Historical launch arguments are not reconstructed from this default alone.

Outcome semantics and attribution. The September 9 extract contains fifteen records labeled as opponent wins. The previous trainer did not parse that label and skipped them. The revised reader preserves losses, draws, and unknowns distinctly. It deduplicates stable game identities, quarantines contradictory records, and excludes redundant match summaries when game records are available.

Publicly visible cards may have been discarded, milled, or exiled. Moreover, GRE action lists can describe offered actions rather than executed plays. The revised extractor retains visibility observations without calling them played-card evidence. Chronological proximity to a draft is also only a candidate game-to-draft association. Training requires a verified or explicit link; current timeline-only records are excluded.

Training status. The gameplay script is now described as outcome-weighted imitation. By default, a verified draft/seat receives weight $0.5+\bar{r}$, where known wins, losses, and draws contribute 1, 0, and 0.5 to $\bar{r}$. This retains losing examples without attributing every loss to a particular card. Global winning-card bonuses and generated alternative-pick rewards default to zero. Reported likelihood is a training diagnostic. No new model has been trained from the repaired data path in the results below.

5. Evaluation Protocol

We distinguish four questions: whether the policy imitates a logged choice; whether it follows a generated teacher; whether its score responds to a relevant context change; and whether following its recommendations improves gameplay. Only the first three have measurements here, with different levels of evidence.

5.1. Historical experiment records

We retain two archived evaluations: a full human-pick split reported as 228,715 states, and a separate 20,000-state human-pick sample. Their preference-accuracy columns evaluate value-generated preferences, not the human labels or observed game wins. We reproduce the numerical tables from saved CSVs; the underlying training runs are not rerun in this revision. These sets were used in repeated model and hyperparameter selection and should be treated as historical development evidence rather than untouched final tests.

5.2. Current-list review queue

The new queue is pinned to a 540-card local cube snapshot. Its list differs from the saved 5.0 list by 64 replacements. Cube updates alter the availability of interactions, so examples are checked against the pinned list rather than transferred indiscriminately from older Arena or other powered cubes.

The queue contains twelve authored control/treatment pairs (24 cases) and 126 decisions sampled from 135 extracted decisions across three recent drafts. Names are canonicalized, cube membership is checked, and complete pick-prefix lengths are required. Selection of real decisions is deterministic. Each real draft family is assigned wholly to one split; each authored pair likewise remains together. Development has 18 probe states and 82 real decisions from two drafts (40 and 42 selected decisions). The reserved test portion has six probe states and 44 real decisions from the third draft. Only development rankings are evaluated or reported.

The probes were authored during assistant-assisted implementation. They are small diagnostic pool ablations, not sampled complete draft trajectories. Every case is pending independent review, and no acceptable-choice labels have been approved. The test portion is reserved for later use, not described as a validated expert benchmark. A logged human choice remains an observation, not a gold-standard strategic label.

5.3. Baselines

All learned baselines use the repaired inference path and the same pinned cube and fallback settings.

- **Raw policy**: the outcome-weighted checkpoint with value, heuristic synergy, prior, and companion adjustments disabled.
- **Default**: that same checkpoint with the configured card prior and companion rules; value and heuristic synergy remain disabled.
- **Aggressive**: the July 100,000-step game-card DPO checkpoint, with $\lambda = 1$, $T = 1.2016$, prior and companion adjustments, and heuristic synergy disabled.
- **Sparse prior**: only the configured standalone card prior, with alphabetical ties. This is not a comprehensive expert power ranking.

The aggressive comparison changes both checkpoint and value reranking. It is a configuration comparison, not an isolated causal estimate of DPO's effect. The continued-imitation checkpoint in the historical sample is also different from the new raw-policy baseline.

5.4. Metrics and reproducibility

Observed-pick agreement is the fraction of the 82 real development states whose logged choice is ranked first. For each probe pair, the pack is fixed while one pool card changes. Given focal candidate a, reference b, control s_0, and treatment s_1, we compute

$$\Delta = [S(a \mid s_1) - S(b \mid s_1)] - [S(a \mid s_0) - S(b \mid s_0)] . \tag{10}$$

A proposed sign $d \in \{-1, +1\}$ is followed when $d\Delta > 0$; zero counts as no directional response. This avoids requiring a large enough shift to change the top pick, but also means arbitrarily small shifts count. The signs require review and are not labels of optimal play. Score magnitudes are not calibrated or directly comparable across checkpoints.

The manifest fingerprints checkpoints, card data, cube, priors, cache, source, settings, and cases. Two saved CPU replays produce byte-identical ranking files and summaries. The paper's table-generation script checks those files, recomputes development metrics from rankings, verifies dependency fingerprints, and rejects test cases in the replay. This establishes reproducibility for the recorded local configuration, not across arbitrary hardware or dependency versions.

6. Results

6.1. Teacher agreement is not a gameplay result

Table 1 shows that value-oriented configurations increase value-teacher agreement while reducing agreement with logged human picks. In the separate sample (Table 2), the aggressive configuration raises value-teacher agreement from 0.5494 to 0.7262 while human top-1 agreement falls from 0.6240 to 0.5092. The historical game-card training report also records 0.958 agreement on its own generated game-card preference evaluation. That figure uses yet another target and must not be substituted for the table's value-teacher metric.

These results show that optimization changes behavior toward its teacher. They do not establish whether those changes improve finished decks. Human disagreement is also not automatically an error: a stronger drafter may deliberately deviate from a human choice. We therefore use these measurements to identify an objective mismatch, not to infer the direction of a win-rate effect.

6.2. Contextual development diagnostics

The raw and default configurations each match 29 of 82 logged choices; the aggressive configuration matches 21 and the sparse prior matches 20 (Table 3). Equal aggregate agreement does not imply identical rankings. These current-list observations cannot be directly compared to historical 20k or full-split accuracies: the examples, cube, model configuration, and inference path differ.

The raw policy follows five of nine proposed directions, and the aggressive configuration follows four. Table 4 identifies specific review priorities. Replacing a control card with Worldspine Wurm decreases Flash's margin against Counterspell, although Wurm's death trigger provides a reason to reconsider Flash. Adding Archon of Cruelty decreases the relative scores of Reanimate or Entomb against another large target in the corresponding probes. Replacing Archon with legendary Titania increases Persist's margin against Reanimate, despite Persist's nonlegendary-target restriction.

These examples motivate testing compatibility and missing roles explicitly. They do not prove that the proposed sign is correct in every full draft, that a particular final pick is wrong, or that a package-aware replacement would improve outcomes. The static prior's zero changes are a construction check, not nine independent empirical failures. Raw and default share a checkpoint, and unconditional prior terms cancel from the paired difference; their matching directions are not independent replications.

The real decisions come from only two development draft families. We therefore report exact descriptive counts, without treating picks as independent trials or presenting an apparently precise state-level confidence interval. Nine authored pairs also cannot support a broad population estimate of combo competence. Independent labels, more draft families, and uncertainty clustered by draft are required for such claims.

Table 1. Archived July full-split results. Human agreement uses the reported 228,715-state split. The preference column uses a separate value-generated preference target; it is neither human-choice accuracy nor win rate. These archived results are not a fresh run of the repaired September scorer.

Configuration	Human top-1	Human top-3	Human MRR	Value-teacher agreement
Outcome policy	0.6066	0.9017	0.7594	0.5504
Value-conservative	0.6000	0.8970	0.7542	0.6538
Value	0.5942	0.8922	0.7495	0.7196

Table 2. Archived July 20k human-state sample. Preference agreement is measured against the separate value-generated teacher.

Configuration	Top-1	Top-3	Teacher
Continued imitation	0.6240	0.9093	0.5494
Safer game-card DPO	0.6161	0.9048	0.6406
Aggressive game-card DPO	0.5092	0.8411	0.7262

Table 3. Current development replay. Agreement is with 82 observed choices, not expert labels. Direction counts concern nine unreviewed probe proposals. No gameplay metric is measured.

Configuration	Matches	Agreement	Directions
Raw policy	29/82	35.4%	5/9
Default	29/82	35.4%	5/9
Aggressive	21/82	25.6%	4/9
Sparse prior	20/82	24.4%	0/9

6.3. Gameplay evidence remains insufficient

All fifteen September 9 opponent-win labels now appear as losses in the audit. The September 20 extraction contains sixteen known outcomes, but all draft associations remain timeline-only. Consequently, none is training-eligible (Table 5). Exclusion does not mean there were no wins; it means the link needed for attribution has not been established.

Forge simulation artifacts exist, including a 100-draft run with 1,200 match records and 1,170 parsed winners. They do not establish superiority over a fixed baseline under controlled pilots or transfer to Arena. We do not report simulator scores, search latency, qualitative lane consistency, or local game counts as evidence of an improved drafting policy. The 54 passing regression tests support the repaired software contracts, not strategic quality.

7. Design Implications and Future Evaluation

The next model should estimate how a candidate changes a feasible deck plan. We propose a versioned registry of packages with compatible enablers, targets, payoffs, tutors, protection, mana requirements, and fallback uses. A package can require more than two roles; some roles have substitutes, while others are unique bottlenecks. Completion, redundant payoffs, opportunity cost, and the likelihood of finding missing roles should all affect a pick. This registry and a package-aware learned policy are not implemented in the evaluated system.

Table 4. Within-model score-margin changes Δ from Eq. 10. The desired sign is an unreviewed proposal. Default has the same changes as raw on these probes; the sparse prior has zero change by construction.

Probe	Sign	Raw	Aggressive
Flash / Wurm	$+$	-0.142	-0.082
Reanimate / target	$+$	-0.247	-0.389
Entomb / setup	$+$	-0.439	-0.687
Breach / Freeze	$+$	$+0.029$	$+0.037$
Persist / target legality	$-$	$+0.068$	$+0.032$
Green fixing	$+$	$+0.221$	$+0.514$
Control curve	$+$	$+0.754$	$+0.604$
Channel / payoff	$+$	$+0.080$	$+0.044$
Saga / Orb	$+$	$+0.020$	-0.043

Table 5. Outcome-record audits. "Eligible" requires verified draft linkage under the repaired training contract. Counts are not policy win-rate estimates; the rows are separate extracts and are not pooled.

Extract	Records	Wins	Losses	Eligible
September 9 legacy	25	10	15	0
September 20 extraction	16	9	7	0

For example, Titania and Zuran Orb form a finite land-sacrifice payoff; recognizing the interaction does not make Orb universally desirable. A reanimation plan may need setup rather than another large creature. A Pact/Oracle plan depends on final-deck construction, including repeated basic-land names. These are candidates for reviewed cases, not unconditional bonus rules. A revised deckbuilder must preserve the interactions valued during drafting and enforce usable mana and curve requirements.

The development sequence is to independently review contextual cases, establish valid outcome linkage, implement a transparent package-aware reranker and compatible deckbuilder, and then learn conditional value. Ablations should separate package features, deckbuilding, priors, and checkpoint changes. Temporal and cube-version holdouts should accompany draft-grouped splits. Preference distillation should follow an independently validated teacher rather

than serve as its evaluation.

Search should be revisited only after its evaluator passes these tests. A recorded sequence of later packs is not a faithful counterfactual once an earlier model pick differs: that pick can alter what other seats receive and select. Sequential evaluation needs an explicit draft/opponent model or prospective play. Gameplay comparison must record model configuration, recommendation adherence, final decks, pilots, opponents, and result identities; control the pilot/deckbuilder where possible; and prespecify effect sizes and stopping rules. None of that prospective comparison is claimed here.

8. Limitations and Conclusion

Our strongest evidence is a reproducible account of what the current system computes and how it responds on a small development set. Historical experiments were repeatedly used for selection; current probe directions are unreviewed and assistant-authored; observed choices are not optimality labels; and observed games lack verified recommendation attribution. The main policy does not directly encode general card rules, the old partial-pool value path remains available experimentally, and no package-aware replacement has yet been tested. We report neither expert-rated improvement nor a causal gameplay effect.

The audit shows why these distinctions matter. Contextual inputs can coexist with context-independent preference supervision, a learned deck evaluator can be misapplied to incomplete pools, and telemetry defects can invalidate outcome feedback. Repairing those contracts and separating evaluation targets provides a defensible foundation for the next drafting model. It does not itself establish stronger play.

Impact and Data Statement

The intended use is draft decision support. Recommendations can influence entry-fee spending, so unvalidated scores should not be represented as proven gameplay advantages. Public-data manifests attribute the source to 17Lands and record its license. Local gameplay logs and checkpoints are not asserted to be publicly released. This working paper reports aggregates and card-level diagnostic examples without publishing account or match identifiers. Independent human expert review of the new cases is pending.

References

17Lands. 17Lands public datasets. https://www.17lands.com/public_datasets, 2026. Dataset provenance is additionally recorded in the local release manifests.

Barash, Z. Arena powered cube draft. https://magic.wizards.com/en/news/mtg-arena/arena-powered-cube-draft, September 2026. September 8, 2026 revision; accessed September 20, 2026. The publisher updates this page in place.

Bertram, T., Fürnkranz, J., and Müller, M. Predicting human card selection in Magic: The Gathering with contextual preference ranking. In *IEEE Conference on Games*, 2021. URL https://arxiv.org/abs/2105.11864.

Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C. D., and Finn, C. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, 2023. URL https://arxiv.org/abs/2305.18290.

Scryfall. Scryfall bulk data. https://scryfall.com/docs/api/bulk-data, 2026. Card metadata source; the evaluated local cache is fingerprinted in the experiment manifest.

Zaheer, M., Kottur, S., Ravanbakhsh, S., Póczos, B., Salakhutdinov, R., and Smola, A. J. Deep sets. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://arxiv.org/abs/1703.06114.

A. Artifact Ledger and Reproduction

The paper's numerical tables are generated by `paper/build_evaluation.py`. Its provenance output records SHA-256 input hashes, verifies identical saved development replays, and checks the scorer and model dependencies against their recorded fingerprints. Missing or changed evidence causes validation to fail. Historical summaries and current rankings remain separate artifacts.

Evidence	Artifact relative to `mtga_cube_draft/`
Current cases and split	`benchmarks/powered_cube_2026_09/cases.jsonl` and `manifest.json` in that directory
Development ranks and settings	`reports/contextual_baselines_2026_09_20/`
Replicated development output	`reports/contextual_baselines_reproduced/`
Historical full-split sweep	`reports/full_eval_value_conservative_sweep_2026-07-01.csv`
Historical 20k sample	`reports/prefacc_500k_20k_sweep_2026-07-02.csv`
Legacy outcome audit	`reports/gameplay_audit_2026_09_09_legacy.json`
New outcome audit	`reports/gameplay_audit_2026_09_20.json`
Implementation notes	`docs/redesign_milestone1.md`

The cube hash is `b8ffcf0c75a386535aeca005d1fa372eb11906fa5c7ed880971b137ba5081e7e`; the cases hash is `673dcc06620038f082f4817123fb0322bbb944f965a1b11b254df4177a0bf705`. The raw/default checkpoint hash begins `9e37335a4bae`; the aggressive checkpoint hash begins `2a6430dd4209`. Full hashes and effective settings are in the replay manifest. The static comparator does not use neural logits even though the evaluation harness loads the common assistant object.

To validate and rebuild paper tables, run from the workspace root:

```
mtga_cube_draft/.venv/bin/python paper/build_evaluation.py
```

To reproduce inference, run from `mtga_cube_draft/`:

```
.venv/bin/python scripts/evaluate_contextual_benchmark.py \
  --verify-manifest reports/contextual_baselines_2026_09_20/manifest.json \
  --out reports/contextual_baselines_paper_replay
```

The default split is development. No command used for this paper requests the reserved test split. Compilation instructions and the distinction between local validated artifacts and portable manuscript files are in `paper/README.md`.

B. Historical Pair-Effect Experiment

Table 6. Archived aggregate outcome evaluation for card-main-effect and pair-interaction fits. These are descriptive selected-run results, not independent evidence against the strategic importance of combinations.

Variant	Game-weighted log loss	Aggregate squared error
Card-only	0.67798	0.05528
1k pairs	0.67943	0.05594
3k pairs	0.68319	0.05763
1k pairs, stronger penalty	0.67929	0.05588

The implementation's metric named `brier` is $\sum_j n_j(p_j - \hat{p}_j)^2 / \sum_j n_j$, where $\hat{p}_j$ is the observed build-level success rate. It is not the ordinary per-game Brier score: the latter adds the within-build term $\sum_j n_j \hat{p}_j(1-\hat{p}_j) / \sum_j n_j$. The same dataset permits comparison of the aggregate squared errors, but their absolute values should not be presented as per-game probability accuracy.

The 1k-pair regularized fit used a minimum of 200 co-occurring games; the 3k-pair fit used 100. These fits had worse selected evaluation loss than the card-only fit. That result rejects neither nonlinear interactions nor structured packages. Sparse observations, confounding by player and archetype, model capacity, and regularization remain alternative explanations. No uncertainty interval or fresh held-out confirmation was produced for this comparison.

C. Review Contract

Cases record source, family, split, full decision context, rationale, and review status. Observed choices remain separate from approved labels. Reviewers must identify themselves, allow defensible alternatives, and resolve compatibility, mana, and missing-role questions independently of the old teacher. Ambiguous proposals should be revised or excluded. Intentional label changes require a new fingerprint and evaluation version.